LA-ICP-MS Analysis with

Signal Integration for Laboratory Laser Systems

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Introduction

Welcome to SILLS, a laser ablation ICP-MS data processing package based on MATLAB. This software package applies to the analysis of any material, with special attention to geological materials, such as minerals, fluid inclusions and melt inclusions. This version of SILLS has several new features:

- Graphical visualization of instrument drift
- Deconvolution of mixed signals (especially applicable to inclusion analysis and $Li_2B_4O_7$ fusions)
- Quantification based on internal standard concentrations, salinities, or wt.% oxides.
- Error calculations based on counting statistics and ICP flicker noise.
- A tool to help eliminate spikes.
- Up to 7 standard measurements including linear drift correction.
- Results preview and custom report output features.
- The ability to display signal ratios, especially for zircon dating
- Multiple file load for decreased data reduction time
- Set integration intervals automatically to reduce data reduction time
- A standalone version without the need of MATLAB is also available

We hope you judge SILLS easy to use and helpful for your LA-ICP-MS application.

Operating requirements

1. MS Excel (for the production of output tables)
2. MATLAB 7 or higher for the script version

Warning: Text and textboxes may be truncated if a low-resolution monitor is used.

Installation

1. Script version
   - Extract the file to a folder of your choice, e.g. C:\Program Files\Matlab\SILLS
2. Standalone version with MATLAB Component Runtime (MCR)
   - Run the file SILLS_pkg_MCR.exe
   - A DOS Command prompt will appear
   - The MCR installation should begin automatically (after some time), if not, run the Windows Installer file
   - As soon as you run SILLS.exe for the first time, SILLS is installed automatically
3. Standalone version without MATLAB Component Runtime (MCR)
   - Install this version only when you have MCR installed.
   - Run the file SILLS_pkg.exe
   - A DOS Command prompt will appear
   - As soon as you run SILLS.exe for the first time, SILLS is installed automatically

Format of the data files

SILLS uses the ELAN system output files as standard input files. An example is provided in the "Extras" folder. Note the following points:

- The first line can be filled with an arbitrary description, but must not be empty
- The first element of the second line has to be "Time in Seconds"
- The isotopes must be called for example **Li6**, NOT **6-Li** or any comments in brackets as **Li6 (HR)**
- Delimiters are commas
- The script version includes the script convert.m which converts data of an ELEMENT-2 instrument. Feel free to modify this script for your instrument.
- You are welcome to test other data formats, but we do not guarantee any functionality

Launching SILLS and working with projects

Launching SILLS

1. Script version
   - Open MATLAB
   - Make sure variables of scripts you used before are saved or can be discarded because the workspace will be cleared launching SILLS
   - Find the SILLS directory in the 'current directory' section at the top of the MATLAB window (Fig. 1)
   - In the command window, type >> SILLS (Fig. 1)
2. Standalone version
   - Run the SILLS.exe file by double-clicking or calling it from a DOS command prompt
   - Do NOT close the appearing DOS command prompt, all unsaved work will be lost

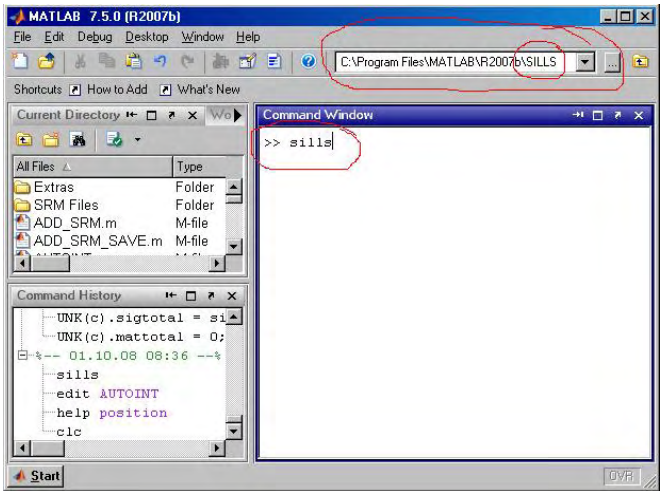


Fig. 1: The MATLAB platform

- The SILLS control panel will appear (Fig. 2):

Saving projects

At any point, you can opt to save your SILLS session by clicking *File* → *Save Project*. You will be prompted to enter a file name, and the file will be stored in .mat format.

Opening projects

If you wish to open a previous SILLS project, click on *File* → *Open Project* and find the appropriate file. You will first be given the option to save the current project.

Note: If you open an old project, all MATLAB figure windows will be closed. Only one SILLS project can be worked on at a time.

Note 2: If errors occur opening a project, try restarting SILLS.

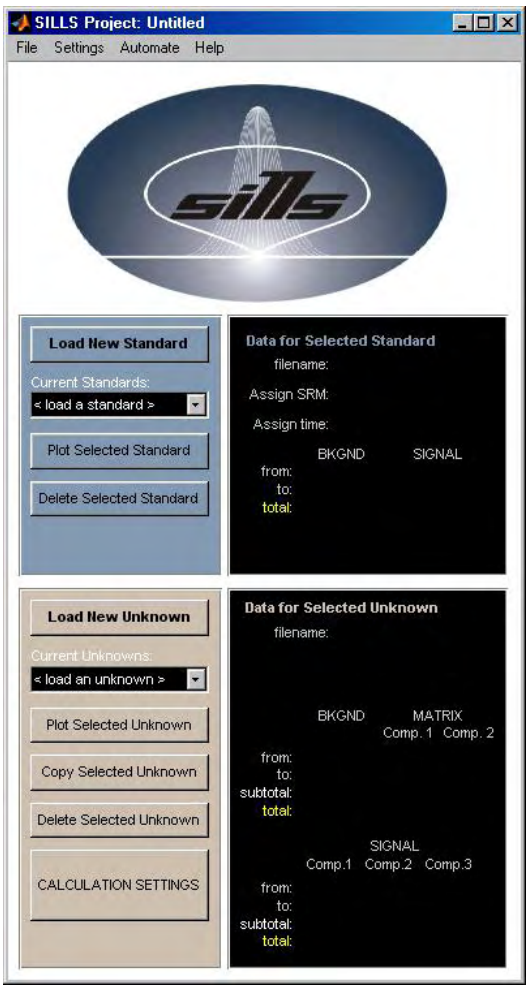


Fig. 2: The SILLS Control Panel

Exiting SILLS

Click on *File* → *Exit* or cancel the SILLS control panel. You will be prompted to save the current session.

Note: If you close the MATLAB main window (Fig. 1) you will also close SILLS. You are prompted to save or exit SILLS. If you erroneously closed the Matlab main window choose yes and cance and you can continue working.

Settings and SRM's

Loading and defining Standard Reference Materials (SRMs)

The 'SRM Files' subdirectory contains .xls files containing concentration data for SRM glasses 610, 612, 614, and 616 from the National Institute for Standards and Technology (NIST) Values. These concentrations are the preferred averages quoted by Pearce et al. (1996). The values for NIST 610 are from the list at the end of this manual. To define your own SRM compositions, do the following:

1. Go to *Settings* → *Standard Reference Materials* → *Define New SRM*
2. A new window (Fig. 3) will appear, into which concentrations for any element can be entered (in µg/g).
3. Next click on *Save SRM to File* and specify a file name. The new SRM will be stored as a .xls file which you can retrieve by clicking *Settings* → *Standard Reference Materials* → *Load SRM from File*
4. *You can change existing values using excel*
5. *You can create SRM files directly with excel similar to existing files.*

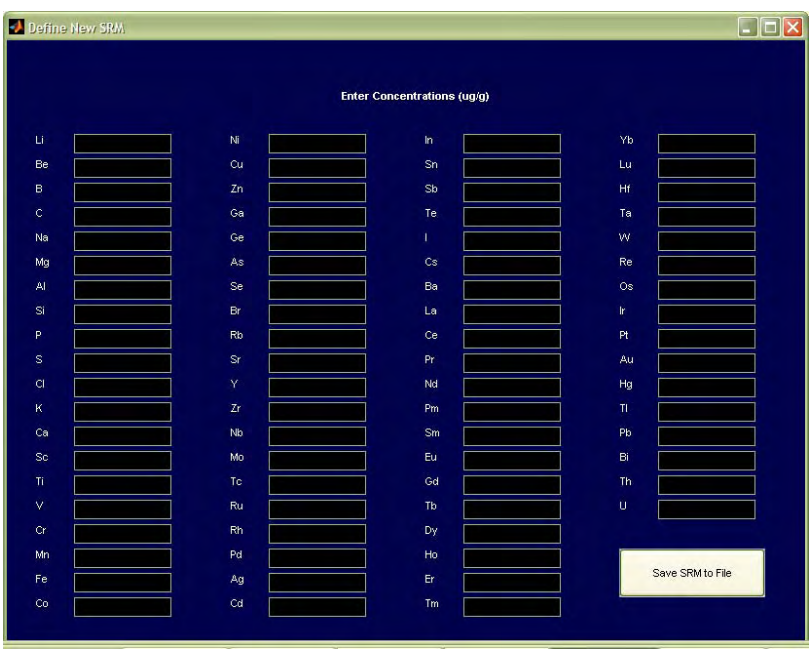


Fig. 3: Window for defining new standard reference materials

Defining the input format

SILLS accepts .csv input files with time in the leftmost column and isotope count rates (or raw counts) in the remaining columns. Columns must have headers labelled with the relevant isotope (e.g. 'Na23'). In order to specify whether the input data is in counts per second or absolute counts, go to *Settings* → *Input Format* and click on either *Counts per second* or *Absolute counts.* The format of these files is described earlier in this manual.

Defining the time format

SILLS is able to calculate instrument drift over time, based on the time your standards were collected. Times can be entered in two formats: either real clock time (hh:mm) or as integer "time points". The default setting is hh:mm format. To specify times as time points,

1. Go to *Settings* → *Time Format*
2. Click on *Integer time points*
3. A separate window will appear, in which you can enter the total number of time points.

Note: You can swap between hh:mm and time point format at any point, and the data for each will not be lost.

Defining dwell times

Once you have loaded at least one standard or unknown, you need to specify the dwell times of each analyte.

1. From the SILLS control panel click *Settings* → *Set Dwell Times*. The following will appear (Fig. 4):

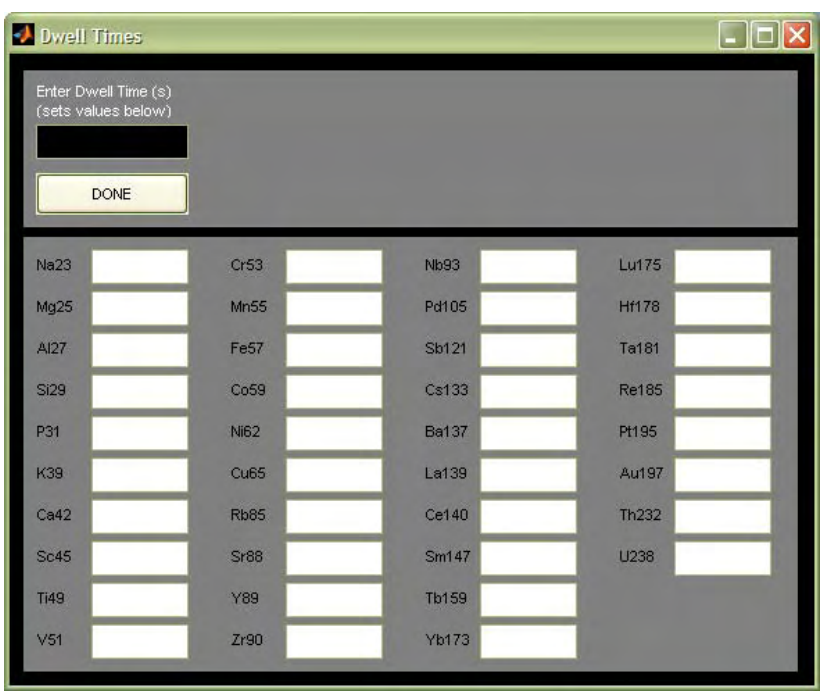


Fig. 4: Window for entering dwell times.

2. Enter the dwell time values (in seconds) in the white box next to each isotope.

Note: To save time, enter the common (or most common) dwell time in the black box at the top of the window.

3. Click on *DONE*.
4. This procedure is also necessary when the input file is in counts per second cps. Reason is the LOD calculation where it is possible that during the gas blank interval (background estimation) the std is 0 and no LOD can be calculated. In this case the std of the background is calculated (best possible estimate) based on one count measured in the interval.

Note: To increase the speed of data reduction, the default value of the dwell time is set to 0.01 s (10ms)

Defining flicker error

There is natural uncertainty in an ICP-MS signal, which can be accounted for with a user-specified 'flicker noise' error. This value is specific to the each experimental protocol. An explanation can be found in Halter et al. (2002).

1. Click on *Settings* → *Set Flicker Noise*. The following will appear Fig 5:

Fig. 5: Window for defining flicker noise.

2. Set the % flicker noise
3. Define the dwell time for which the % flicker noise was determined (***Note:*** Flicker noise is independent of the element).

It should be mentioned here that the uncertainty of an analysis is described best by multiple analysis of the same sample and calculating the mean and standard deviation of the individual results and not the error given based on flicker noise and counting statistics!!!!!

Loading measurements and plots of measurements

Loading standard measurements

1. Click on *Load New Standard*
2. Select the appropriate file(s). You can load multiple standards (see Loading multiple standards). Your standard will appear in a separate window (Fig. 6).
3. Do the spike elimination (suggested) if required by clicking the spike elimination button. Decription see: Spike Elimination
4. Select integration windows for the background and signal by clicking and dragging across the graphic display. The radio buttons labelled *BKGND* and *SIGNAL* specify which will be selected.
5. You can switch the display mode from intensities to ratios. Description see: Ratio Mode

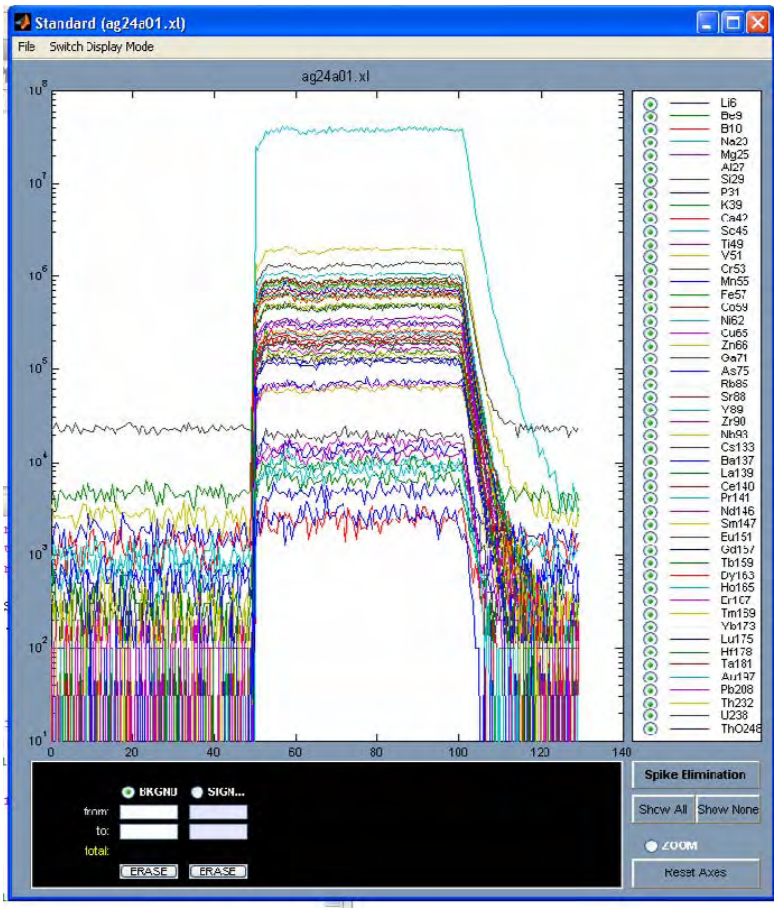


Fig. 6: The standard window

Zooming:

To zoom in on the plot, click *ZOOM* and drag the mouse across the desired region. To restore the original view, click on *Reset Axes*.

Defining time intervals manually:

Instead of using the mouse, specify the boundaries of your integration intervals in the text boxes at the bottom of the standard window.

Viewing individual Elements:

You can look at specific elements by selecting/deselecting the appropriate radio buttons in the figure's legend. Clicking on Show All or Show None does as you would expect!

Erasing integration windows:

If you wish to remove an integration window, click on the appropriate *ERASE* button.

Once background and signal windows have been selected, you can close the standard window. The time intervals are stored as variables in the MATLAB workspace. In the SILLS control panel, you will notice the "Current Standards" list has been updated, and the panel to the right is showing the time intervals you just selected.

Print or save plots as images:

Use the "File" menu to print or save the figure. Various image formats are available.

Additional measurement settings

1. Assign your standard an SRM identity from the *Assign SRM* pop-up window. If the appropriate SRM is not in this list, then either define it or load it from a file (see "Loading and Defining Standard Reference Materials")
2. Specify the actual clock time or time point the standard was analyzed or get the times by retrieving the time automatically from file info.

Note: The information displayed in the SILLS control panel always refers to the current standard.

You can now load additional standards (up to 7), or proceed to loading your unknowns.

Loading unknowns

Loading unknowns is identical to loading standards, except you are given additional options. In addition to the background window, you can select up to 3 separate windows for you sample signal, and up to 2 windows for the matrix (a.k.a. host). In the example in Figure 7, the green window defines the matrix signal, and the blue window defines the sample signal contaminated by the matrix. As for standards, you can eliminate spikes, print/save the figure or switch to the ratio mode.

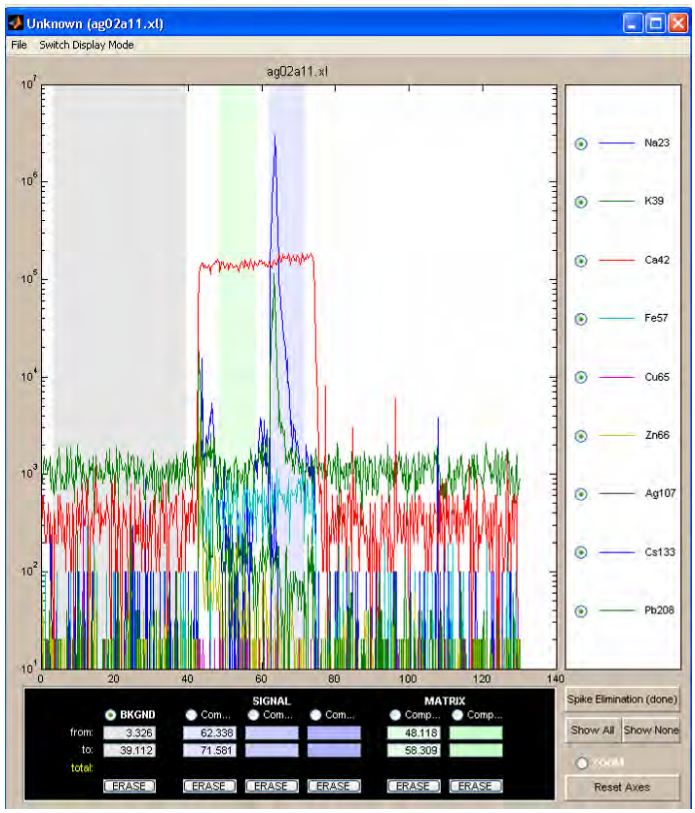


Fig. 7: The unknown window.

Note: In the SILLS control panel, you are not given the option of defining any addition data for the unknown (e.g. clock time). These are all entered separately in the SILLS Calculation Manager.

Continue loading unknowns as necessary.

Deleting standards and unknowns

If you wish to delete a standard or unknown, select the appropriate file from the drop-down list in the SILLS control panel, and click on *Delete Selected Standard* or *Delete Selected Unknown*.

Plotting standards and unknowns

If you wish to view a standard or unknown that you have closed, click on *Plot Selected Standard* or *Plot Selected Unknown*. You can then edit integration windows, etc.

Copying unknowns

If it is more convenient to copy an existing unknown than load it again, click *Copy Selected Unknown*. The copy is treated as an entirely separate unknown, and you can modify its settings as you wish.

Spike elimination (based on Grubbs test)

An algorithm to identify and correct possible spikes or outliers has been implemented in the software. The whole dataset with intensities above a selectable threshold (e.g., 1000 cps) (Fig. 8) is tested for outlying points by the method described by Grubbs (Grubbs 1969). This test assumes a normal distribution, and compares the calculated statistic test value (STV, equation 1 below) with a tabled value depending on the number of measurements (7 in our case) and level of significance (1%, meaning the detected values are highly significant outliers):

$$STV = \frac{|x^* - \bar{x}|}{s} \quad (1)$$

x^*: outlier suspected value

$\bar{x}$: mean, calculated with all values in the series

s : standard deviation, calculated with all values in the series

When outliers are detected, the software asks the user to confirm the spike and to confirm that the outlier should be replaced with the calculated, suggested value (mean of the measurements before and after the detected spike), or another user defined value, or to keep the original value. Once a spike is corrected, the original values can only be recovered by reloading the original data file, which will remain unchanged as SILLS continues to be used. Modification of outlier data is a controversial practice questioned by many scientists, including the authors. The Grubbs test algorithm provides an objective method for spike identification, and the user has to confirm the modification for every single detected spike. This does not make the practice scientifically or methodologically valid. The rejection of outliers is more acceptable when it is done based on an underlying model or when the possible source of the spike is confidently known, e.g., contaminant particles accidentally mobilized into the ICP–MS from the transport or ablation system.

Fig. 8: Pop up window of the spike elimination

If a possible spike is found the following window pops up (Fig. 9). In the center the identified spike is graphically shown (fat blue line with dot) and a correction value is suggested (fat black). You now have the choice to correct, enter a custom value, ignore the spike or finish the spike elimination. In the case of Fig. 9 I would correct the value. When spike elimination has checked all values, a summary appears as shown in Fig. 10. Close this window.

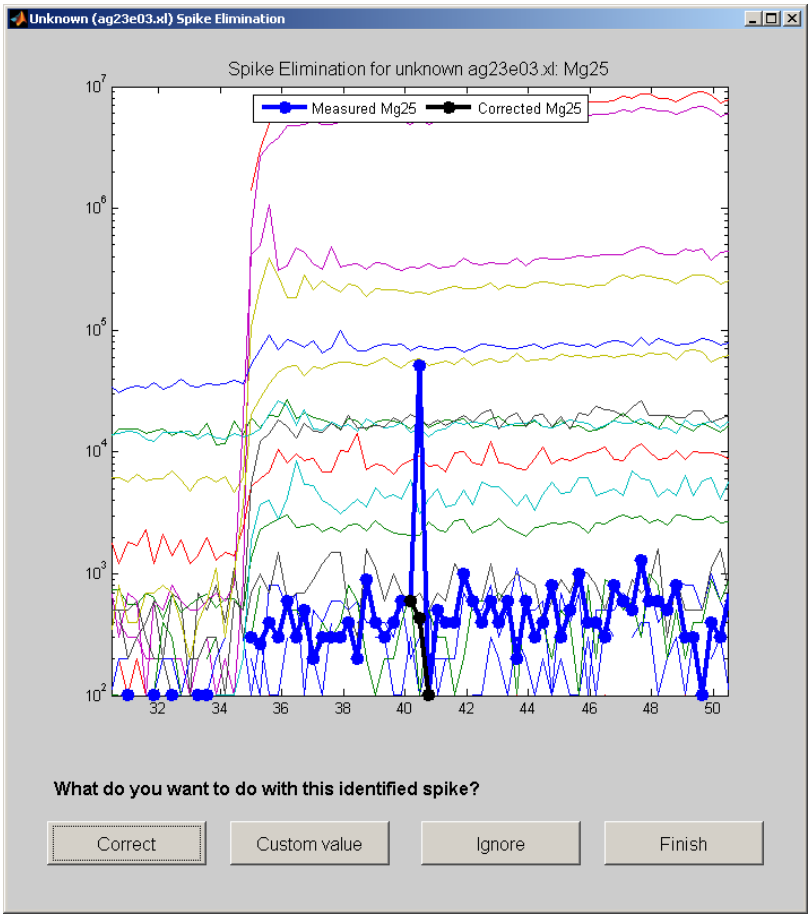


Fig. 9: Spike elimination window

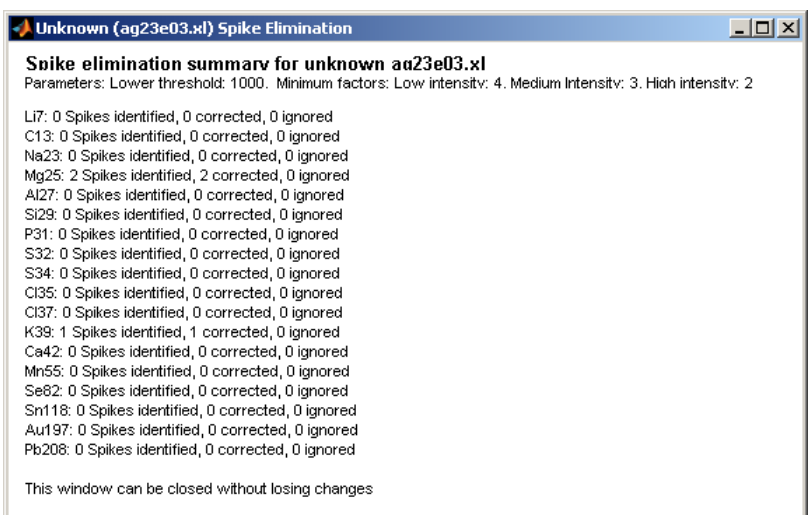


Fig 10: Spike elimination summary

Ratio mode

To make signal integration easier for certain applications (i.e. zircon dating) it is possible to switch the display mode to ratios using the "Switch Display Mode" menu (Fig. 7). Note that you have to define the background integration window first, because the displayed ratios are background-corrected. First, a window (Fig. 11) will pop up where you can define the ratios by selecting the elements from the lists. Check the "SHOW" boxes for the ratios you want to have displayed in the signal integration window (Fig. 12), the other ratios will be calculated and appear in the results. Confirm clicking "DONE". You can switch back to intensities or redefine the ratios using the "Switch Display Mode" menu again.

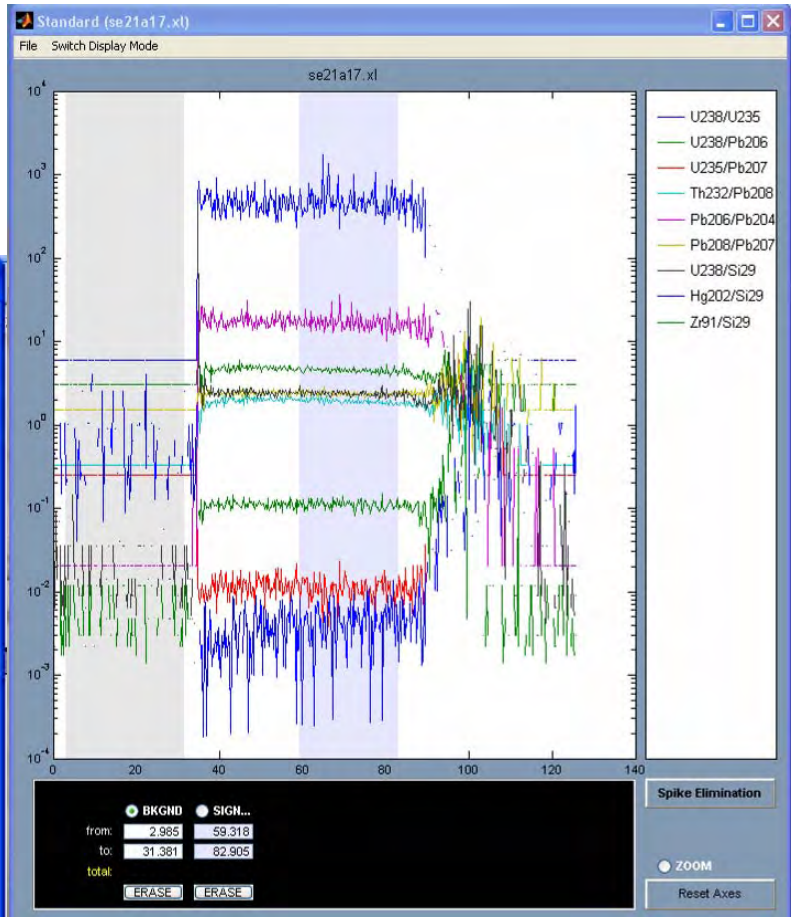


Fig. 11: Ratio Definition

Fig. 12: Ratio display mode

Note: If you switch between intensity, spike elimination and ratio figures multiple times it can happen that some error appears. In that case, close the figures and plot the standard or unknown again from the SILLS control panel.

Automation:

Loading multiple standards

It is possible to load multiple standards by clicking on load new standard and than use the control key to select all the standard measurements at once (Fig. 13). They then are directly loaded without plotting them, so no integration intervals are set. It is possible to set integration intervals automatically. (See setting integration intervals automatically)

Note: To have the standards in the "right" order (in the drop down menu of the standards) choose the last standard measured first and go to the first measured (a wrong order has no influence on the result since each standard needs a time.)

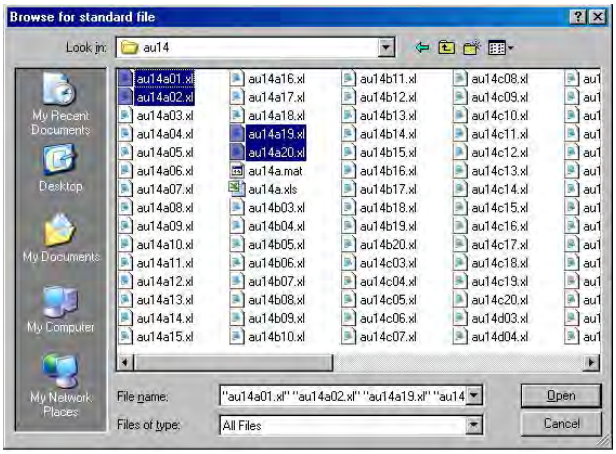


Figure 13: loading multiple standards by control click.

Loading multiple unknowns

It is possible to load multiple unknowns by clicking on load new unknown and than use the shift and/or control key to select all the standard measurements at once (Fig. 14). They then are directly loaded without plotting them, so no integration intervals are set. It is possible to set integration intervals automatically. (See setting integration intervals automatically)

Note: To have the unknowns in the "right" order (later in the calculation settings and the results file) choose the last unknown measured first and go to the first measured (a wrong order has no influence on the result since each standard needs a time.)

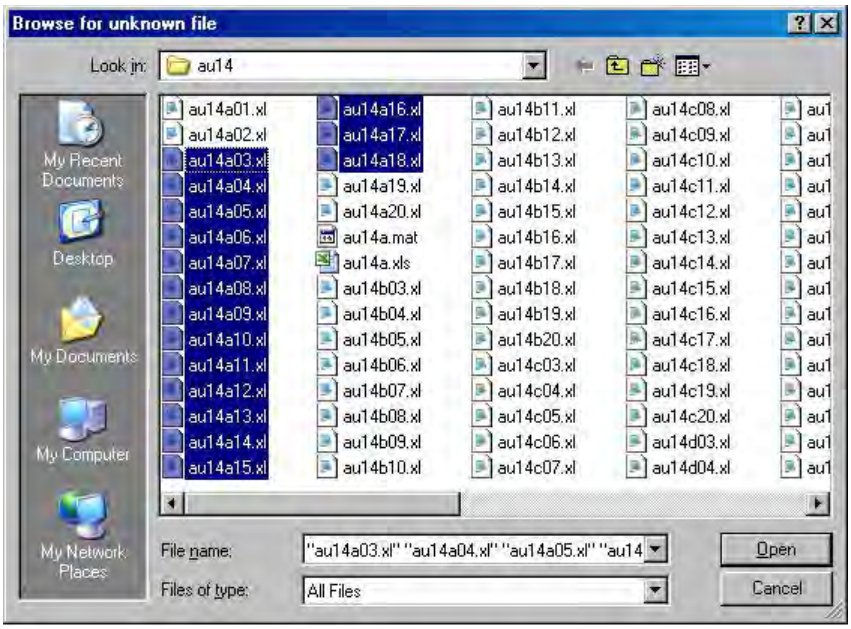


Figure 14: loading multiple unknowns by control and/or shift - click.

Retrieve the time automatically from the file info

To retrieve the time automatically from the file info go in the SILLS main window to automate/ retrieve Times from file info. A dialog will appear: "This option retrieves the times of the measurements from the time the data file hase been created. Already entered times will be kept, Time format will be changed to real clock time. If you load additional data files from now on, repeat this command before starting calculations. It is recommended to check the times for plausibility."

Click on yes and check the retrieved times for plausibility.

Set integration intervals automatically

To increase the speed of data reduction it is possible to define the integration intervals (Background and signal) automatically. After loading several standards and unknowns (loading multiple standards / unknowns) in the main SILLS window under Automate/Set integration intervals automatically the fallowing window appears (Fig. 15): You can choos the Element based on which the signal start and end is searched (default is the first element measured). You can choose the delay (in time steps not seconds) from the automatically found signal start to the signal integration begin to exclude any possible surface contamination or fractionation effect at the beginning of an ablation. (The default value is 10 time steps). You can choose the length of the signal after the delay either constant for all measurements by entering the number of time steps in the "length of the signal after delay" box or leave the default value of zero, then the software automatically tries to find the end of the ablation.

This function should be only used for homogeneous solids and integration intervals should be checked for plausibility!

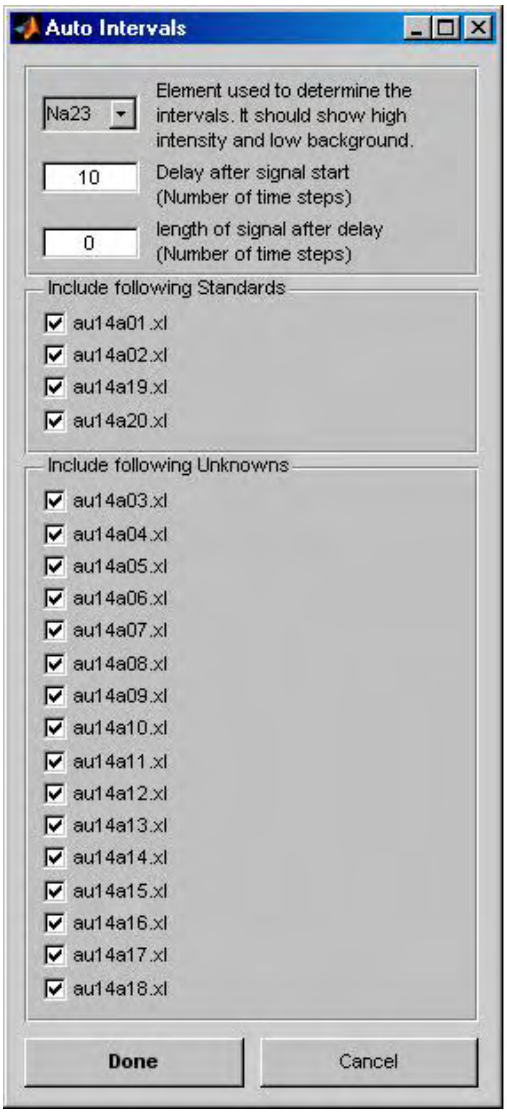


Figure 15: The automatic integration dialog window

Calculation Manager

Specifying Calculation Settings

The possibilities of calculations you can make and constraints you have to set are shown in Fig. 16.

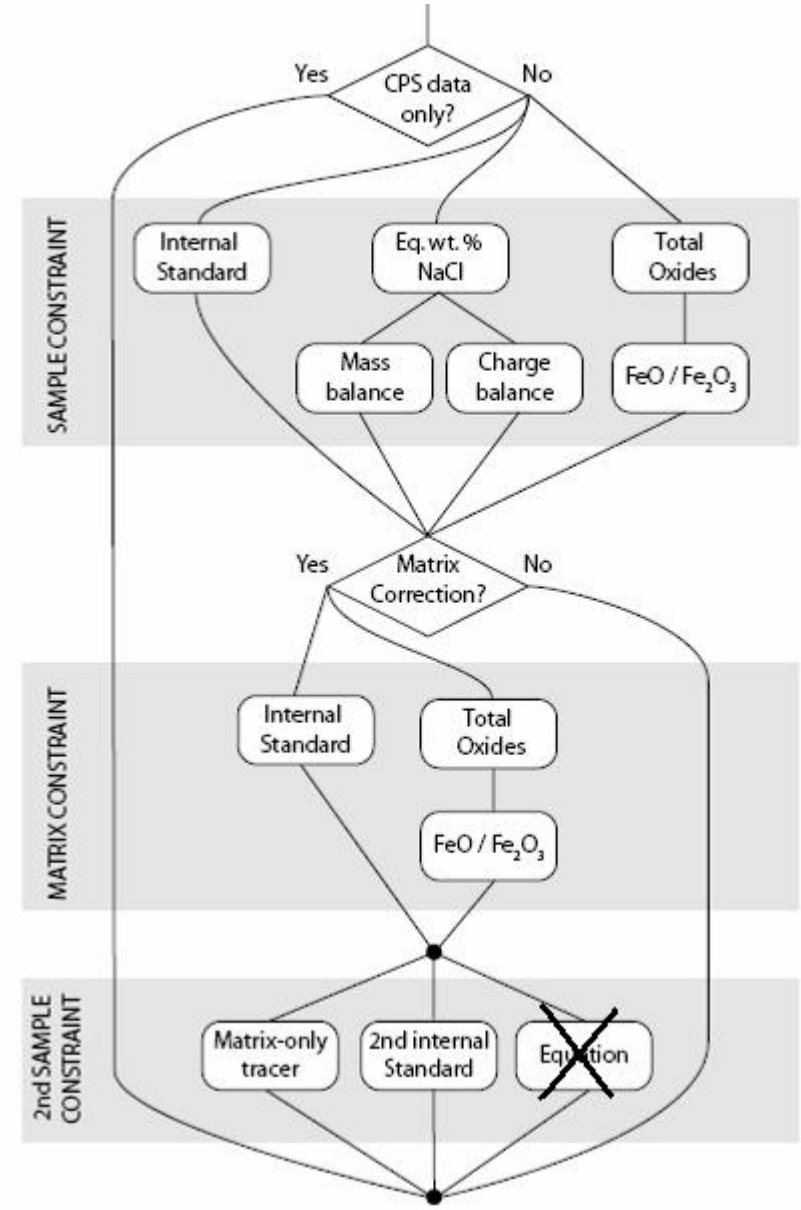


Fig. 16: Possibilities of calculations

Once all standards and unknowns have been loaded and all integration intervals selected, click on *CALCULATION SETTINGS*. A separate window will appear (Fig. 17):

Fig. 17: Calculation Manager window. Here all calculation settings are defined.

You will see a separate row of options for each unknown loaded. On the right side of the page you have the option to *Plot*, *Copy*, or *Delete* unknowns as you choose. On the left side of the page, there is an information box for each unknown, in which you enter any descriptive details for the particular analysis.

Assining times to unknowns:

Under the 'Assign Time' header, either enter the appropriate hh:mm time or select the appropriate time point from the drop-down list. Use the possibility to retrieve the time info automatically.

CPS data only:

Check this box if you don't need any concentration calculations. This applies for example for zircon dating, where only the ratios of the cps are of interest. The Matrix and Sample settings columns will disappear and a button where you can once again redefine your ratios will replace them. (Fig. 18)

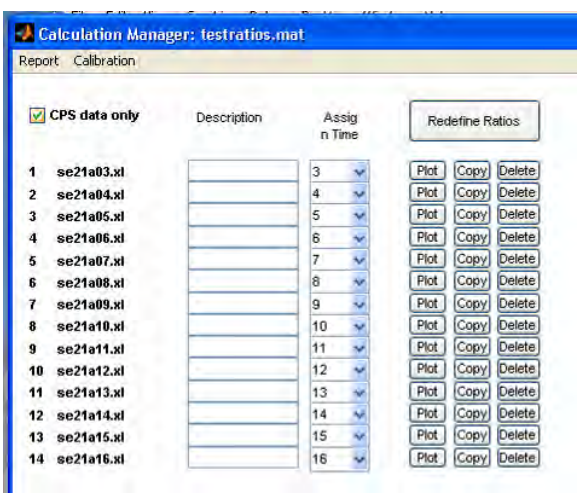


Fig. 18: Calculation Manager window with CPS data only

Matrix quantification settings:

If you have selected any matrix integration windows, the drop-down lists under the 'Correction Type' heading in the green panel will contain the following options:

- **none** (EXPLANATION: No matrix correction applied. The corresponding signal window will be treated as a pure sample signal.)
- **internal std** (EXPLANATION: The composition of the matrix is calculated according to a fixed internal standard concentration.)
  1. Select the file from which you wish to apply the matrix correction (this may be any unknown in which a matrix integration window was selected).
  2. Specify the concentration unit for the internal standard (either *µg/g* or *wt.%*)
  3. If you have selected *µg/g*, specify the internal standard isotope from the list.
  4. If you have selected *wt.%*, specify the internal standard oxide from the list.
  5. Enter the concentration of the internal standard under the 'Value' heading.
  6. If you wish to apply the matrix calculation settings of the one unknown to the entire batch, click on *Apply to All*.
- **total oxides** (EXPLANATION: The composition of the matrix is calculated according to the total wt.% oxides of the major elements (SiO_2, TiO_2, Al_2O_3, Fe_2O_3, FeO, MnO, MgO, CaO, Na_2O, K_2O, and P_2O_5).

  1. Select the file from which you wish to apply the matrix correction (this may be any unknown in which a matrix integration window was selected).
  2. Enter the total wt.% oxides for the matrix under the 'Value' heading.
  3. If you have measured an isotope of Fe, enter the $FeO/(FeO+Fe_2O_3)$ ratio.
  4. If you wish to apply the matrix calculation settings of the one unknown to the entire batch, click on *Apply to All*.

Sample quantification settings (without matrix correction):

If you have selected the matrix correction option 'none', you will see quantification settings for the various unknowns in the SILLS Calculation Manager window. A series of options appear under the 'Correction Type' heading:

- **internal standard** (EXPLANATION: The composition of the sample is calculated according to a fixed internal standard concentration.)
- **wt.% NaCl (mass)** (EXPLANATION: If you have analysed Na, you have the option of calculating the composition of the sample according to the equivalent wt.% NaCl value, using a mass balance expression of the type:

  $[NaCl]_{equiv} = [NaCl] + A \cdot \Sigma[X_iCl_{ni}]$

  where A is a weighting factor and X_iCl_{ni} are auxiliary chloride salts and is set per default to 0.5

  1. Enter the equivalent wt.% NaCl value under the 'Value' heading
  2. Click on *ELEMENTS*
     a. A separate window will appear (Fig. 19):

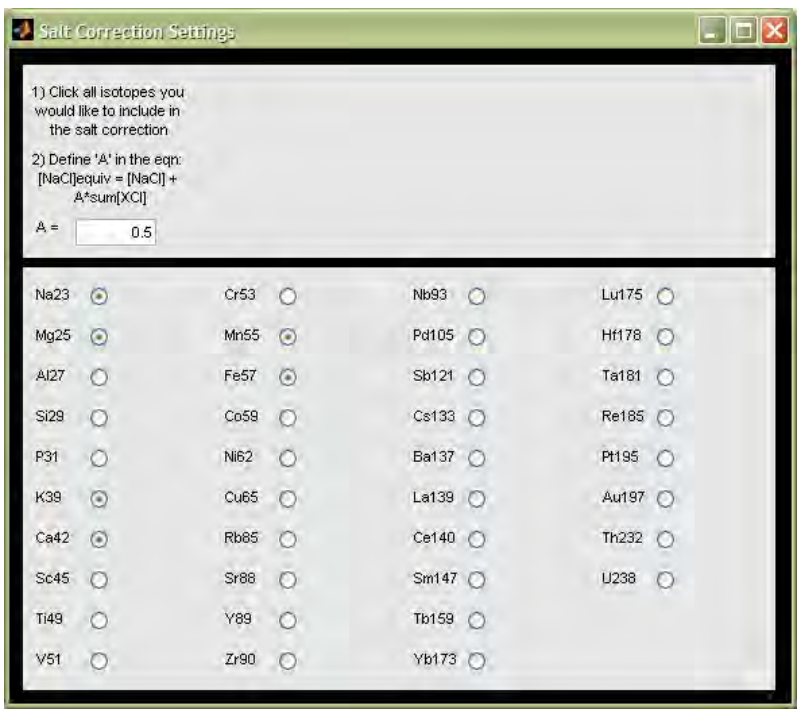


Fig. 19: Salt Correction Settings

     b. Click on all elements you would like to include as auxiliary elements in the mass balance expression.

     Note: If you have analysed multiple isotopes of the same element, click on just one for the salt correction. Otherwise, the salt correction will overestimate the effect of that element.

c. Define the weighting factor 'A' in the mass balance expression.

d. Close the 'Salt Correction Settings' window.

5. If you wish to apply the sample calculation settings of the one unknown to the entire batch, click on *Apply to All*.

- **wt.% NaCl (charge)** (EXPLANATION: If you have analysed Na, you have the option of calculating the composition of the sample according to the equivalent wt.% NaCl value, using a charge balance expression of the type:

  $[Na] = [Na]_{equiv} + [1 + \Sigma([X_iCl_{ni}] / [NaCl])]^{-1}$

  1. Enter the equivalent wt.% NaCl value under the 'Value' heading.
  2. Click on *ELEMENTS*, and proceed as instructed.
  3. If you wish to apply the sample calculation settings of the one unknown to the entire batch, click on *Apply to All*.

- **total oxides** (EXPLANATION: The composition of the sample is calculated according to the total wt.% oxides of the major elements (SiO_2, TiO_2, Al_2O_3, Fe_2O_3, FeO, MnO, MgO, CaO, Na_2O, K_2O, and P_2O_5).

  1. Enter the total wt.% oxides for the sample under the 'Value' heading.
  2. If you have measured an isotope of Fe, enter the $FeO/(FeO+Fe_2O_3)$ ratio.
  3. If you wish to apply the matrix calculation settings of the one unknown to the entire batch, click on *Apply to All*.

Sample quantification settings (with matrix correction):

If you have opted for a matrix correction (either by internal standard or total oxides), do the following:

1. Click on *DEFINE SAMPLE SETTINGS*. The following window will appear (Fig. 20):

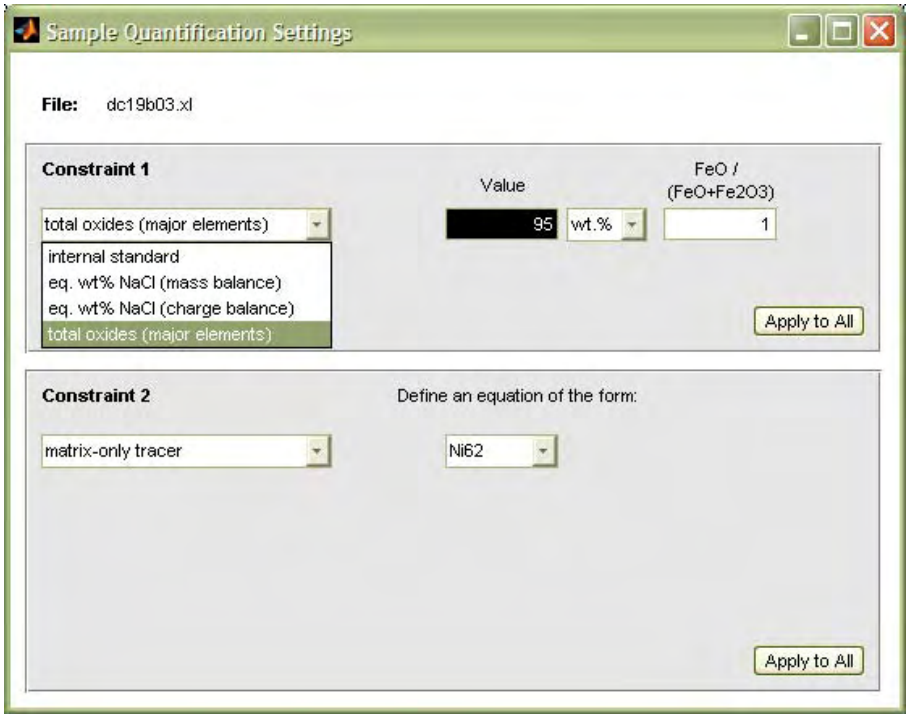


Fig. 20: Sample quantification settings window

2. Under the heading 'Constraint 1' define the composition of the sample by *internal standard, wt.% NaCl (mass balance)*, *wt.% NaCl (charge balance)*, or *total oxides (major oxides)* as above. Click on *Apply to All* if you wish these settings to be applied to all samples.
3. Under the heading 'Constraint 2', you have the following options:
   a. **matrix-only tracer** (EXPLANATION: if a given element is absent from the sample but occurs in the host, you can define it as a 'matrix-only tracer' from the drop-down list.)
   b. **2nd internal standard** (EXPLANATION: if there are two internal standards defined for the sample, define the second internal standard's concentration here).

4. Again, if you wish to apply the settings of 'Constraint 2' to all samples, click on *Apply to All*.

Drift correction and visualization of the instrument drift

For instrumental drift correction, visualization and quality control based on the standard reference material measurements do the following in the SILLS Calculation Manager:

1. Click on *Calibration* → *Show Drift*. A window such as the following(Fig. 21) will appear:

For the drift correction you have to define the element, relative to which the drift correction is done (red marked in Fig. 17). This element is assumed to undergo no drift during an analysis run. It is set to the first (lightest measured) element but usually should be set to the element used as internal standard

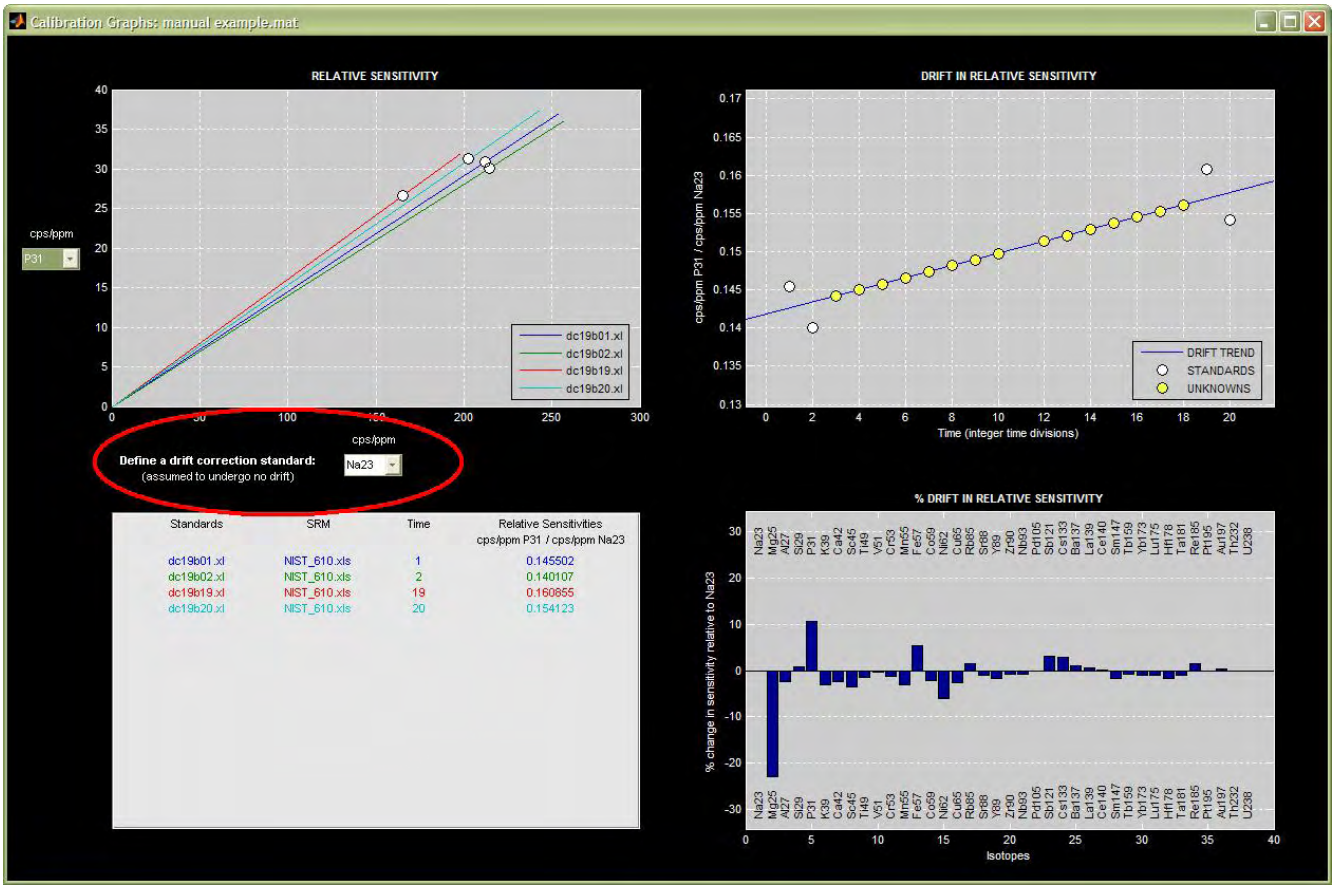


Fig. 21: Calibration Graphs window

PLOT 1: RELATIVE SENSITIVITY: The graph in the top-left compares the sensitivity of any two elements, i.e. plots cps/ppm for isotope A vs. cps/ppm for isotope B. A separate line is shown for each standard measured. Compare the relative sensitivity of different isotopes by selecting them from the drop-down lists.

PLOT 2: DRIFT IN RELATIVE SENSITIVITY: The graph in the top-right shows the relative sensitivity of isotope A and isotope B as a function of time (as specified by the user). White dots represent the standards and yellow dots represent the unknowns. The slope is the least-squares regression line through the standard data, with each standard weighted equally.

PLOT 3: % DRIFT IN RELATIVE SENSITIVITY: The bar graph in the bottom-right shows the percentage drift in each isotope (based on the calculated drift between the first and last measured standard).

Quality control:

Depending on your application, system, SRM, measured isotopes and many other parameters you have to judge if your quantification is valid or if there was a problem. "Normally" Drift should not exceed 5% for most elements if element is homogeneous distributed in the SRM. If there are elements with very high drift you should check the transient signal of the SRM measurement for spikes, remove spikes or possibly one of the SRM measurements.

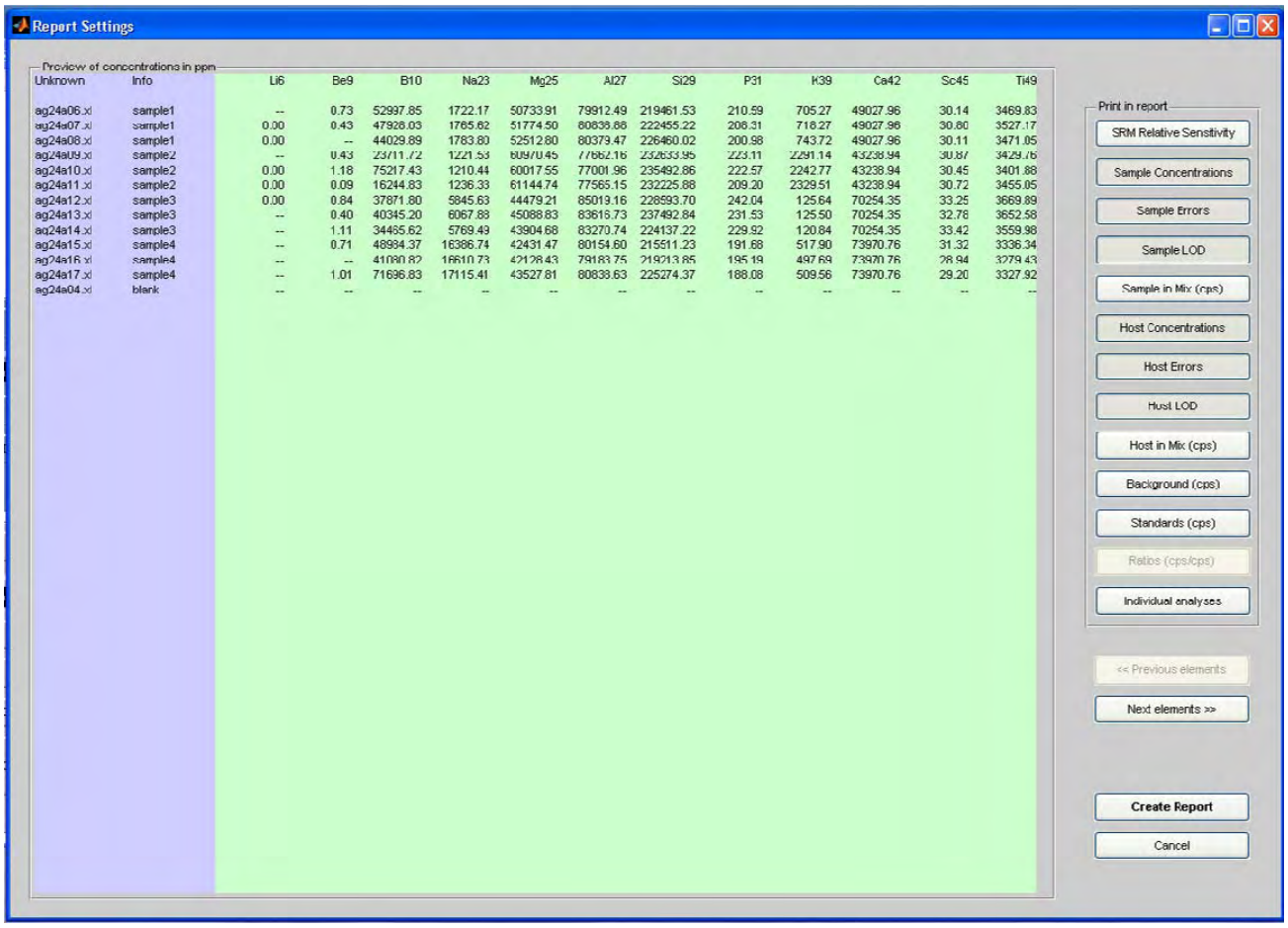
Creating an output report

Define report settings:

1. If you wish to view major elements in the output as wt.% oxides, go to the SILLS Calculation Manager and click on *Report* → *Settings* → *Major Elements as Oxides*.
2. To specify which isotope you would like ratio expressed to in the output report, click on *Report* → *Settings* → *Show Ratios to Which Element?* → [select isotope]
3. Set LOD filter factor: The LOD calculation formulae contain a fixed factor 3. You can change this factor, for example if all your measurements are below detection limit but you wish to see them anyway. It is recommended not to change this for regular measurements.

Create a report:

1. Click *Report* → *Create Output Report*
2. A window like Fig. 22 appears including a preview of the (ppm) concentrations of the unknowns for the first 12 measured isotopes. You can look at more elements by using the "next elements >>" button. From the right side it is possible to choose, what will be saved in a final report to excel. Finally create the report. If you have chosen "CPS data only" the cps will be shown in the preview instead of the concentrations. If you wish to modify the default options, edit the script Report_config.m (script version only)
3. Specify the filename. The output is saved as a *.xls file which can be accessed in MS Excel or similar spreadsheet software.



Unknown	Info	Li6	Be9	B10	Na23	Mg25	Al27	Si29	P31	K39	Ca42	Sc45	Ti49
ag24a06.xl	sample1	--	0.73	52997.85	1722.17	50733.91	79912.49	219461.53	210.59	705.27	49027.96	30.14	3469.83
ag24a07.xl	sample1	0.00	0.43	47928.03	1765.62	51774.50	80838.88	222455.22	206.31	716.27	49027.96	30.80	3527.17
ag24a08.xl	sample1	0.00	--	44029.89	1783.80	52512.80	80379.47	226460.02	200.98	743.72	49027.96	30.11	3471.05
ag24a09.xl	sample2	--	0.43	23711.72	1221.53	60970.45	77662.16	232633.95	223.11	2291.14	43238.94	30.87	3429.76
ag24a10.xl	sample2	0.00	1.18	75217.43	1210.44	60017.55	77001.96	235492.86	222.57	2242.77	43238.94	30.45	3401.88
ag24a11.xl	sample2	0.00	0.09	16244.83	1236.33	61144.74	77565.15	232225.88	209.20	2329.51	43238.94	30.72	3455.05
ag24a12.xl	sample3	0.00	0.84	37871.80	5845.63	44479.21	85019.16	228593.70	242.04	125.64	70254.35	33.25	3669.89
ag24a13.xl	sample3	--	0.40	40345.20	6067.88	45088.83	83618.73	237492.84	231.53	125.50	70254.35	32.78	3652.58
ag24a14.xl	sample3	--	1.11	34465.62	5769.49	43904.68	83270.74	224137.22	229.92	120.84	70254.35	33.42	3559.98
ag24a15.xl	sample4	--	0.71	48984.37	16386.74	42431.47	80154.60	215511.23	191.68	517.90	73970.76	31.32	3336.34
ag24a16.xl	sample4	--	--	41080.82	16610.73	42128.43	79183.75	219213.85	195.19	497.69	73970.76	28.94	3279.43
ag24a17.xl	sample4	--	1.01	71696.83	17115.41	43527.81	80838.63	225274.37	188.08	509.56	73970.76	29.20	3327.92
ag24a04.xl	blank	--	--	--	--	--	--	--	--	--	--	--	--

Figure 22 Results preview and report settings window

Examples and special comments:

XRF of $Li_2B_4O_7$ fusion pill measurements

It is easily possible to correct for the matrix (Li2B4O7) when a blank sample is measured in each run. In the example nr. 3 is the blank value. For the blank measurement you don't need to define a signal window. After all measurements are integrated go to the calculation settings and do the fallowing:

1. Define the internal standard concentrations for the samples, i.e. a major element of the XRF analysis (Fig. 23)
2. Choose for the blank measurement the correction type internal standard and give for Li the value 80500 µg/g (stochiometric calculation of the Li concentration in Lithium tetraborate) (Fig. 24)
3. Choose from the blank measurement to apply the matrix setting to all. (Fig. 24)
4. Check if the concentrations of the internal standard are in the first constraint and the second constraint is matrix tracer only (lithium) (Fig. 25)
5. Proceed to the output report (main elements as oxides and ratio to e.g. calcium)

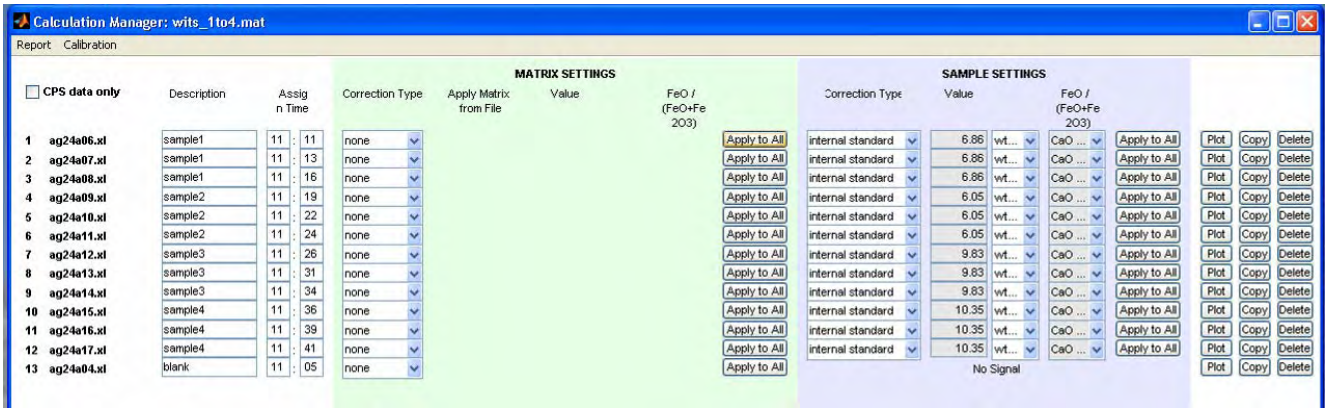


Figure 23

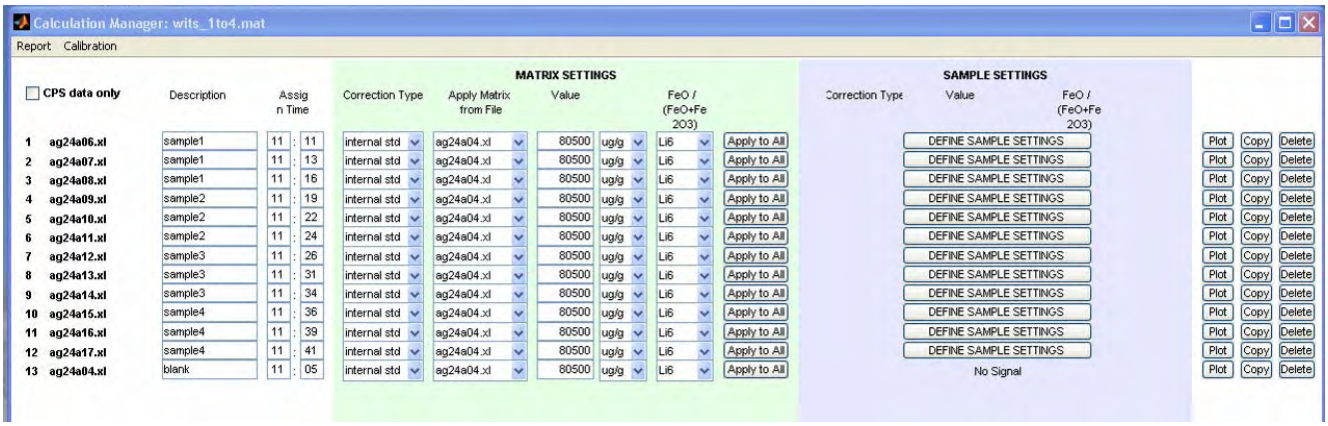


Figure 24

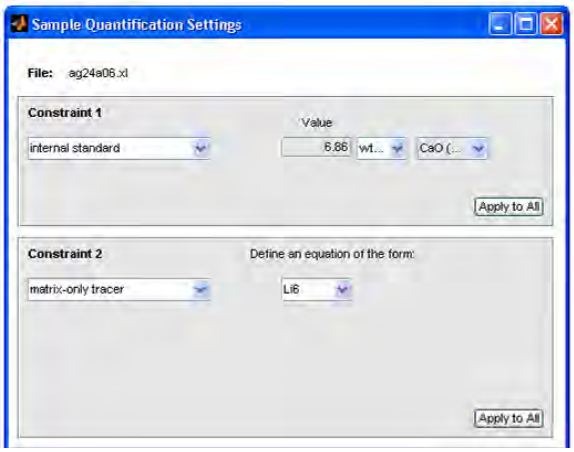


Figure 25

Known problems, bugs and workarounds:

mtimes error (matrix dimensions must agree):

Hit enter in all internal standard concentrations in the calculation manager

Chang from wt% to mg/g problem.

Restart the Project

Futher Information:

References

http://www.igmr.ethz.ch/research/fluids/software

http://www.igmr.ethz.ch/research/fluids/software/sills_folder/MAC_SC_40_Sills_description

NIST 610 concentrations and source

ELEMENT	NIST 610 (wt-% ox.)	NIST 610 (wt-ppm)	
H			
He			
Li		484.6	Pearce et al.1997
Be		465.6	Pearce et al.1997
B		356.4	Pearce et al.1997
C			
N			
O			
F			
Ne			
Na	13.3520	99052.8	Pearce et al.1997
Mg		465.3	
Al	2.0390	10791.4	Pearce et al.1997
Si	69.9750	327090.7	Pearce et al.1997
P		342.5	Pearce et al.1997
S			
Cl		470.0	Pearce et al.1997
Ar			
K		465.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Ca	11.4500	81833.3	Pearce et al.1997
Sc		441.1	Pearce et al.1997
Ti		434.0	Pearce et al.1997
V		441.7	Pearce et al.1997
Cr		405.2	Pearce et al.1997
Mn		433.3	Pearce et al.1997
Fe	0.0588	457.1	Pearce et al.1997
Co		405.0	Pearce et al.1997
Ni		443.9	Pearce et al.1997
Cu		430.3	Pearce et al.1997
Zn		456.3	Pearce et al.1997
Ga		438.1	Pearce et al.1997
Ge		426.3	Pearce et al.1997
As		317.4	Pearce et al.1997
Se		109.0	Pearce et al.1997
Br			
Kr			
Rb		426.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Sr		516.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Y		458.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Zr		437.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Nb		419.4	Pearce et al.1997
Mo		376.8	Pearce et al.1997
Ru			
Rh		1.31	Sylvester and Eggins, 1997
Pd		1.05	Sylvester and Eggins, 1997
Ag		239.4	Pearce et al.1997
Cd		259.4	Pearce et al.1997
In		441.4	Pearce et al.1997
Sn		396.3	Pearce et al.1997
Sb		368.5	Pearce et al.1997
Te		327.0	Jacob, pers. Comm.
I			
Xe			
Cs		357.2	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Ba		454.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
La		440.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Ce		458.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Pr		443.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Nd		436.7	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Sm		453.4	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Eu		444.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Gd		455.5	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Tb		440.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Dy		435.7	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Ho		440.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Er		455.5	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Tm		423.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Yb		455.3	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Lu		439.7	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Hf		421.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Ta		376.6	Pearce et al.1997
W		445.3	Pearce et al.1997
Re		103.7	Pearce et al.1997
Os			
Ir			
Pt		3.15	Sylvester and Eggins, 1997
Au		22.5	Pearce et al.1997
Hg			
Tl		61.2	Pearce et al.1997
Pb		426.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Bi		357.7	Pearce et al.1997
Th		457.0	Rocholl et al., 2000 (TIMS preferred), and Refs therein
Pa			
U		461.5	Rocholl et al., 2000 (TIMS preferred), and Refs therein